

EXAM 7 – SPRING 2014

14. (2 points)

Given the following reinsurance company data (\$000) evaluated as of December 31, 2013:

Calendar/ Accident <u>Year</u>	Earned Risk <u>Pure Premium</u>	Adjusted <u>Premium</u>	Aggregate Reported <u>Loss</u>
2010	5,000	5,000	3,000
2011	4,500	5,500	2,500
2012	5,000	6,000	2,500
<u>2013</u>	<u>6,500</u>	<u>6,500</u>	<u>1,500</u>
Total	20,000	23,000	9,500

Reported Loss Development Factors to Ultimate

12 months	2.50
24 months	2.00
36 months	1.60
48 months	1.25
60 months	1.00

- Adjusted premium incorporates changes in primary rates, underwriting, and exposure affecting the loss potential.

Calculate the Stanard-Bühlmann (Cape Cod) ultimate loss ratio for 2011.

CONTINUED ON NEXT PAGE

SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 14	
TOTAL POINT VALUE: 2	LEARNING OBJECTIVE: A13: Calculate ceded loss reserves using appropriate methods.
SAMPLE ANSWERS	
<i>Sample Answer 1</i> Step 1 – Calculate the reporting lags Report Lags: 2010 = $1/1.25 = 80\%$ 2011 = $1/1.60 = 62.5\%$ 2012 = $1/2.00 = 50\%$ 2013 = $1/2.50 = 40\%$ Step 2 – Calculating the Cape Cod ELR Cape Code ELR = Total <u>Aggregate</u> Reported Loss / sumproduct of <u>Adjusted Premium</u> multiplied by <u>lag pattern</u>) Cape Code ELR = $9,500 / (5,000 \times 0.80 + 5,500 \times 0.625 + 6,000 \times 0.50 + 6,500 \times 0.40) = 72.9\%$ Step 3 – Calculate the IBNR IBNR = $ELR \times ARPP \times (1 - \text{Report Lag})$ 2011 IBNR = $0.729 \times 5,500 \times 0.375 = 1,504$ Step 4 – Calculate the loss ratio 2011 Loss Ratio = $(\text{Aggregate Reported Loss}_{2011} + \text{IBNR}_{2011}) / \text{Earned Risk Pure Premium}_{2011}$ 2011 Loss Ratio = $(2,500 + 1,504) / 4,500 = 89.0\%$	

SAMPLE ANSWERS AND EXAMINER'S REPORT

Sample Answer 2

AY	Adjusted Premium (1)	CLDF (2)	% Reported (3) = 1/(2)	Used Up Premium (4)=(1)×(3)	Reported Loss (5)
2010	5,000	1.25	80.0%	4,000.0	3,000
2011	5,500	1.60	62.5%	3,437.5	2,500
2012	6,000	2.00	50.0%	3,000.0	2,500
2013	6,500	2.50	40.0%	2,600.0	1,500
Total	23,000			13,037.5	9,500

Expected Loss Ratio = Reported Loss / Used Up Premium = 9,500 / 13,037.5 = 72.9%

AY 2011 Ultimate Loss Ratio

= (Reported Loss + Adjusted Premium × ELR × (1.00 - % Reported)) / Earned Risk Pure Premium

= (2,500 + 5,500 × .729 × (1.00 - .625)) / 4,500 = 89.0%

Sample Answer 3

Expected Loss Ratio = Reported Loss / Used Up Premium

= 9,500 / (6,500/2.50 + 6,000/2.00 + 5,500/1.60 + 5,000/1.25) = 72.9%

AY 2011 Ultimate Loss

= Reported Loss + Adjusted Premium × ELR × (1.00 - 1/LDF)

= 2,500 + 5,500 × .729 × (1.00 - 1/1.60)

= 4,002.9

AY 2011 Ultimate Loss Ratio = AY 2011 Ultimate Loss / Earned Risk Pure Premium

= 4,002.9 / 4,500 = 89.0%

EXAMINER'S REPORT

The average score for this question was quite high. Approximately 77% of the candidates received full or nearly full credit.

Common mistakes:

1. Using the wrong premium base when calculating the ultimate loss ratios. Many candidates used the Adjusted Premium instead of the Earned Risk Pure Premium.
2. Candidates calculated the expected loss ratio correctly but then used an expected loss ratio method instead of the Bornhuetter-Ferguson approach (Note: This produces a very similar ultimate loss ratio as the correct answer but it is just by coincidence. $ELR \times \text{Adjusted Premium} = 72.9\% \times 5,500 = 4,007.7$ which translates into an 89.1% ultimate loss ratio $[4,007.7 / \text{Earned Risk Pure Premium}]$.)
3. Candidates calculated an all accident year loss ratio instead of just accident year 2011.